

2. Turn off mains using your main breaker (or wait for a known outage).
3. See what continues running on the inverter.
4. Mark those circuits in your grid as Inverter-backed: Yes/No.

Why this matters: your “brain” devices (router, smart hub/controller) should ideally be on UPS/inverter, otherwise your automations die exactly when you need them most.

Call an electrician for this part

Your DB is where safety and stability begin: MCBs, isolator, and ideally residual current protection.

A simple DB health check (no tools needed)

- Is the DB accessible and not blocked by storage?
- Are breakers labelled (even roughly: “lights”, “kitchen”, “AC”)?
- Is there an RCCB/ELCB/RCBO installed? (a leakage protection device)

If you’re building a modern smart home, leakage protection isn’t optional – especially as you add more powered devices and adapters. Many guidelines and safety references in India strongly emphasize proper protective devices and safe construction/maintenance practices for installations.

Important: Don’t DIY inside the DB unless you’re qualified. This is “licensed electrician” territory.

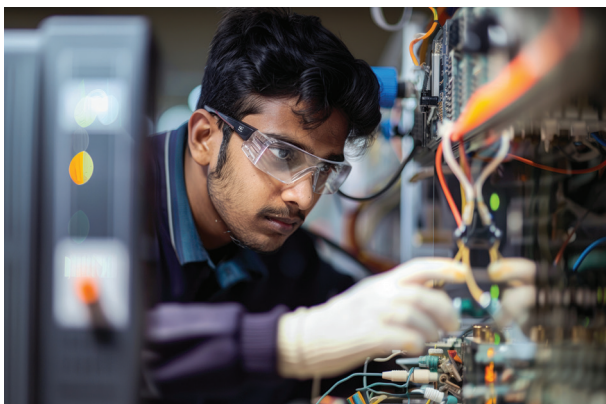
Safety basics

A smart home increases the number of electrical accessories you rely on – plugs, switch modules, extension boards, adapters. That makes standard compliance worth caring about.

What to look for: BIS Standard Mark + licence number

BIS explains that the Standard Mark is used on certified products, and the licence number should be indicated on the product along with the mark (as applicable).

Step-by-step: Verify BIS/ISI authenticity (fast consumer method)



INDIA-FIRST “DON’T DIY” LIST

Automation is fun; mains electricity is not. Do not DIY these unless you’re qualified:

- distribution board work (MCB/RCCB/RCBO changes)
- high-load circuits (geyser, AC lines, kitchen heavy sockets)
- anything in wet areas without proper enclosure/IP rating
- fan regulator rewiring if you don’t understand the circuit

Your goal is a smarter home – not a risky one.

1. Look for the BIS Standard Mark / ISI mark on the product or packaging.
2. Note the licence number (often printed near the mark).
3. Use the BIS CARE app feature “Verify Licence Details” to check authenticity.

Which standards matter most for “home automation builds”?

You’ll commonly encounter these categories under BIS/ISI scope for household electrical accessories:

- **Plugs and socket-outlets:** IS 1293 (household and similar purposes)
- **Domestic switches:** IS 3854 (switches for domestic and similar purposes). BIS even publishes product manuals and implementation notes around revisions.
- **PVC insulated cables:** IS 694 (common for residential wiring).
- BIS’ Scheme-I list includes entries such as IS 3854 (switches) and IS 694 (PVC insulated cables), highlighting their place in the certification ecosystem.

Remember: A smart plug that isn’t built to appropriate standards is worse than a dumb plug – because you’ll trust it to run unattended.

Plan switchboard upgrades without rewiring whole home

You have three practical routes for lighting control in India:

Route A: Smart bulbs (least invasive)

Best for: renters, homes with unknown wiring, quick wins

Downside: wall switch OFF = bulb “dead” (unless you enforce “switch stays ON” habits)

Use smart bulbs when:

- you can’t access wiring safely